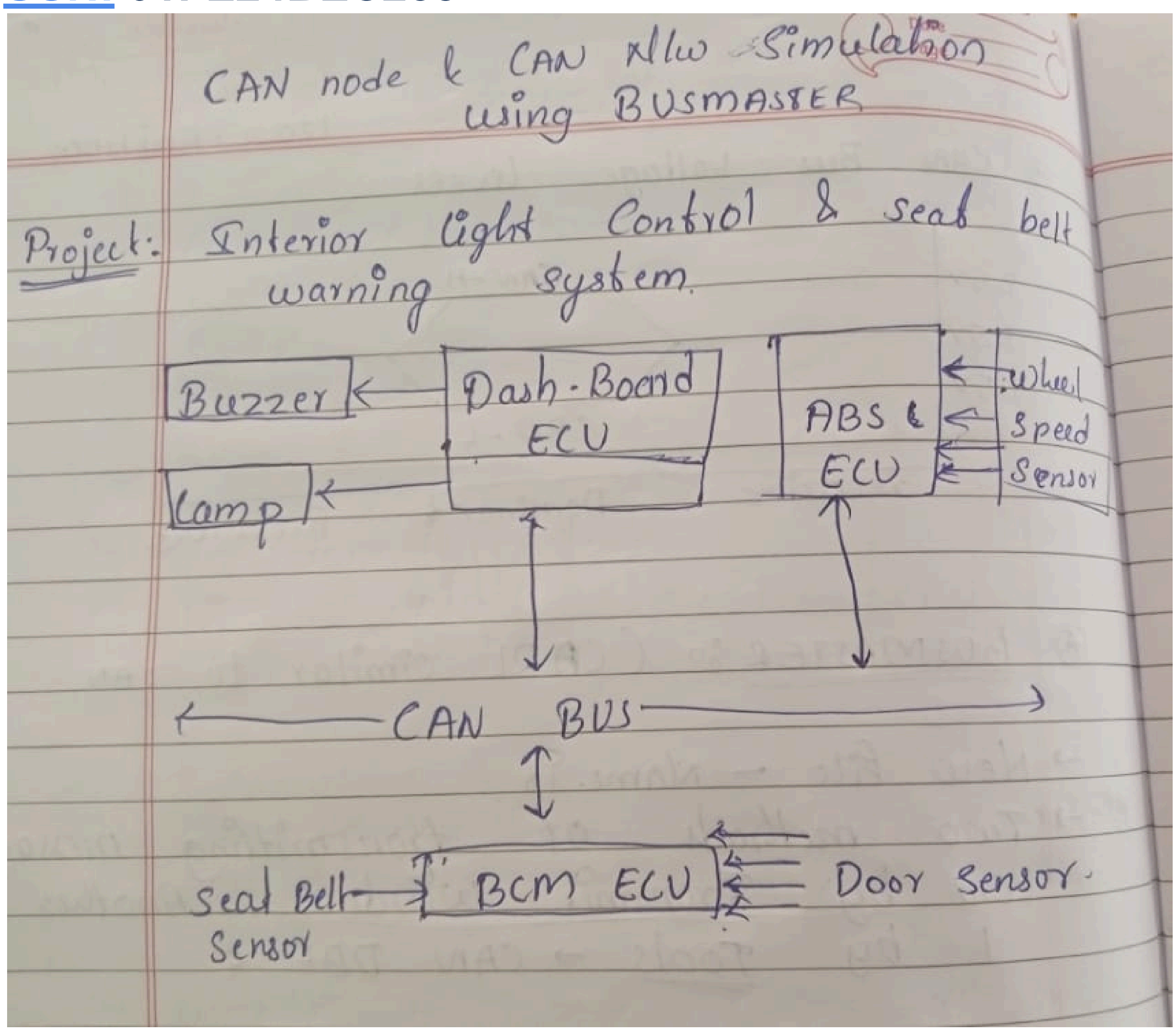
PROJECT on CAN

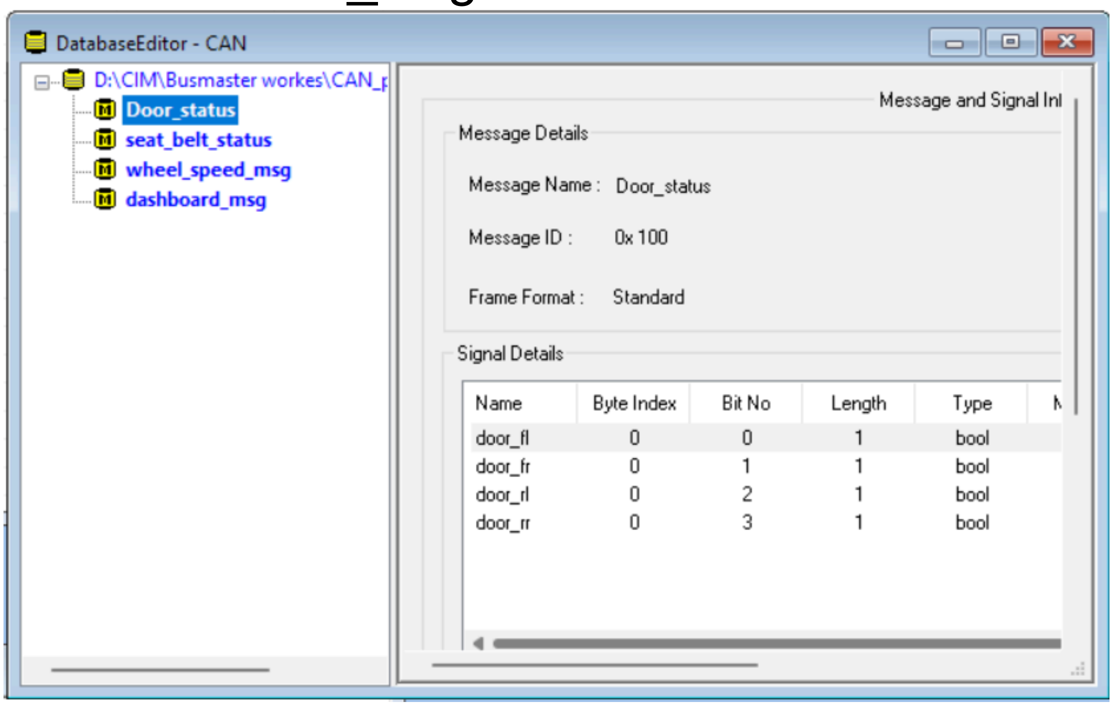
Title: Interior Light Control & Seat Belt warning system.

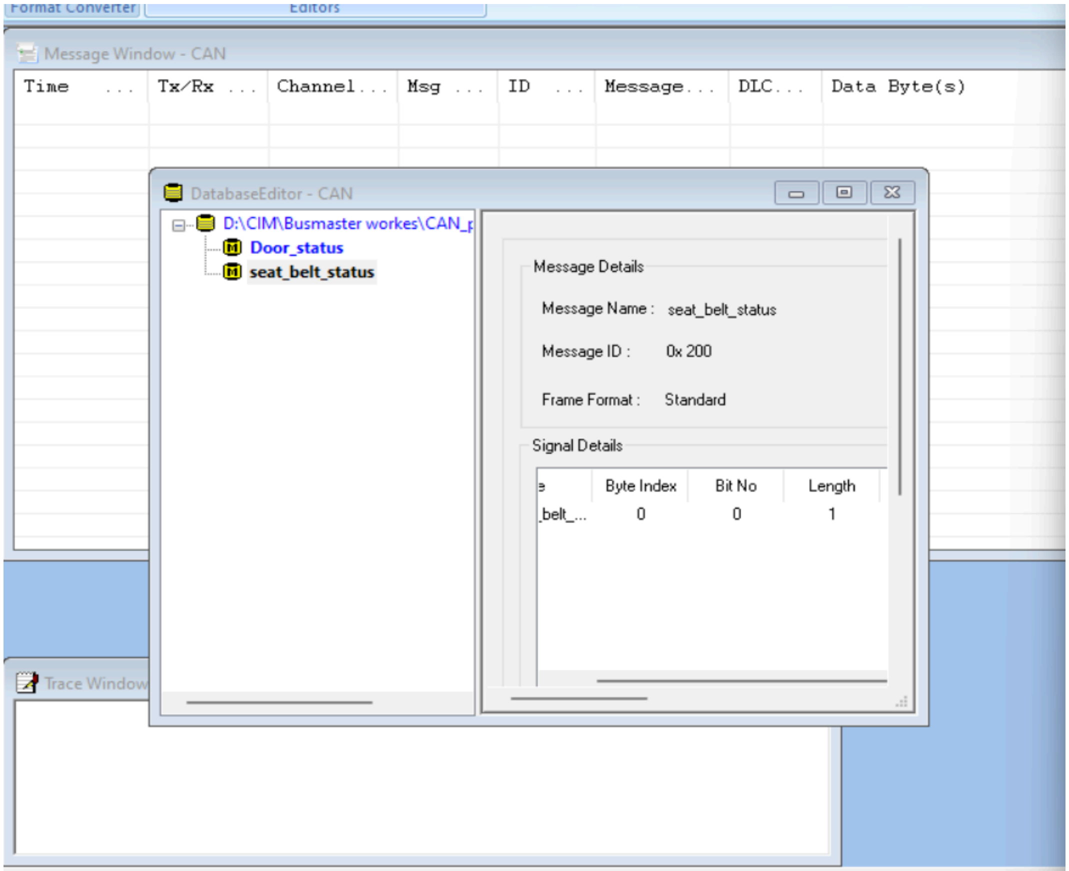
Name: **Chinmay N. Yalawatti**
USN: **01FE24BEC258**

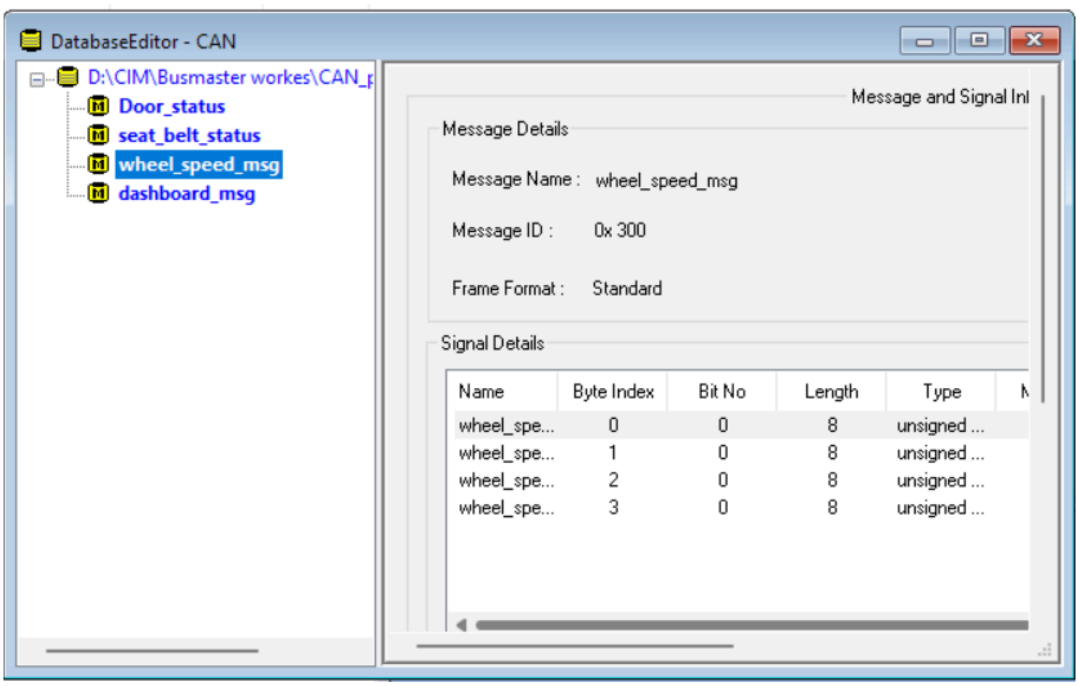


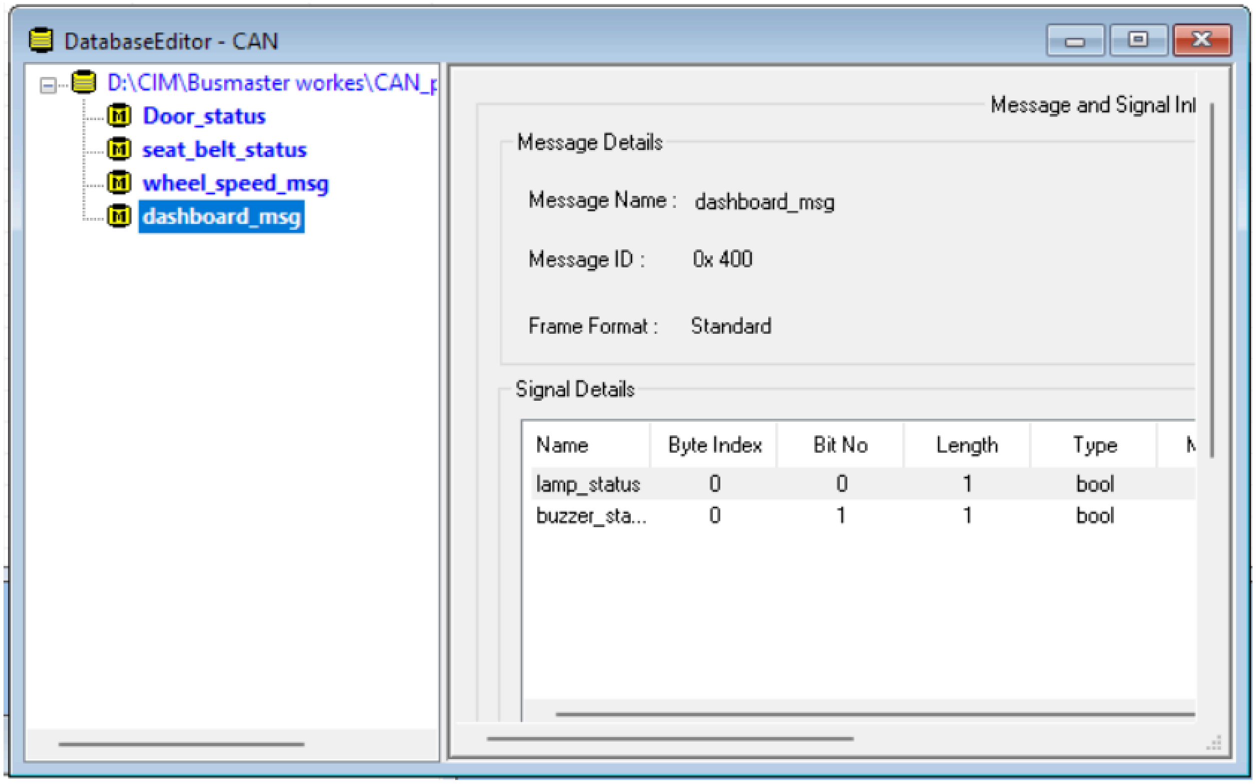
1)Creating Database form CAN -DatabaseEditor

- Creating 4 dataBase
- Door_status
- seat_belt_Status
- dashboard_msg




Format Converter
Editors
Message Window - CAN
Time ...
Tx/Rx ...
Channel...
Msg ...
ID ...
Message...
DLC...
Data Byte(s)
DatabaseEditor - CAN
D:\CIM\Busmaster workes\CAN_p
Door_status
seat_belt_status
Message Details
Message Name : seat_belt_status
Message ID : 0x 200
Frame Format : Standard
Signal Details
Byte Index
Bit No
Length
belt...
0
0
1
Trace Window


DatabaseEditor - CAN
D:\CIM\Busmaster workes\CAN_p
Door_status
seat_belt_status
wheel_speed_msg
dashboard_msg
Message and Signal Inf
Message Details
Message Name : wheel_speed_msg
Message ID : 0x 300
Frame Format : Standard
Signal Details
Name
Byte Index
Bit No
Length
Type
wheel_spe... 0 0 8 unsigned ...
wheel_spe... 1 0 8 unsigned ...
wheel_spe... 2 0 8 unsigned ...
wheel_spe... 3 0 8 unsigned ...

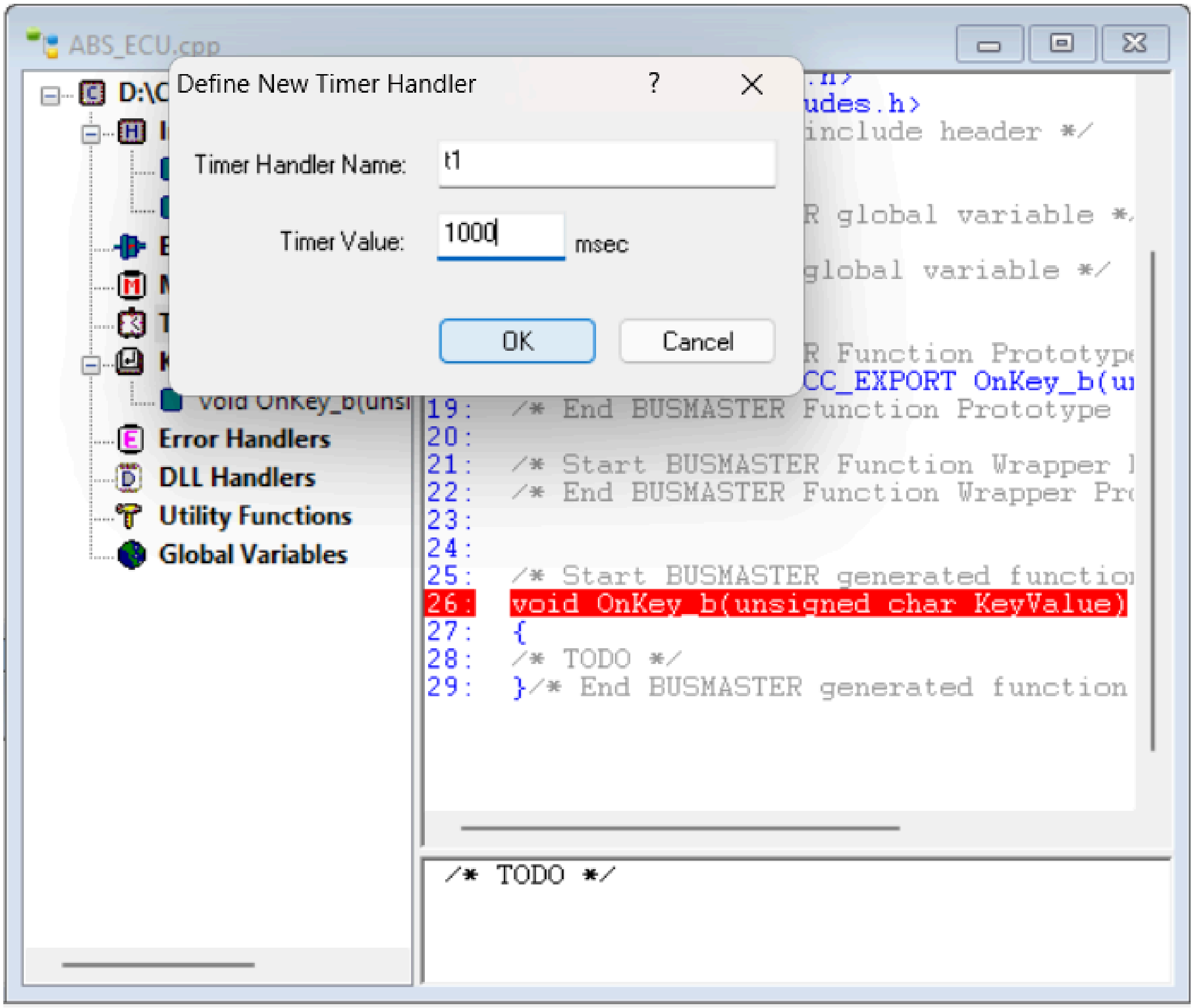


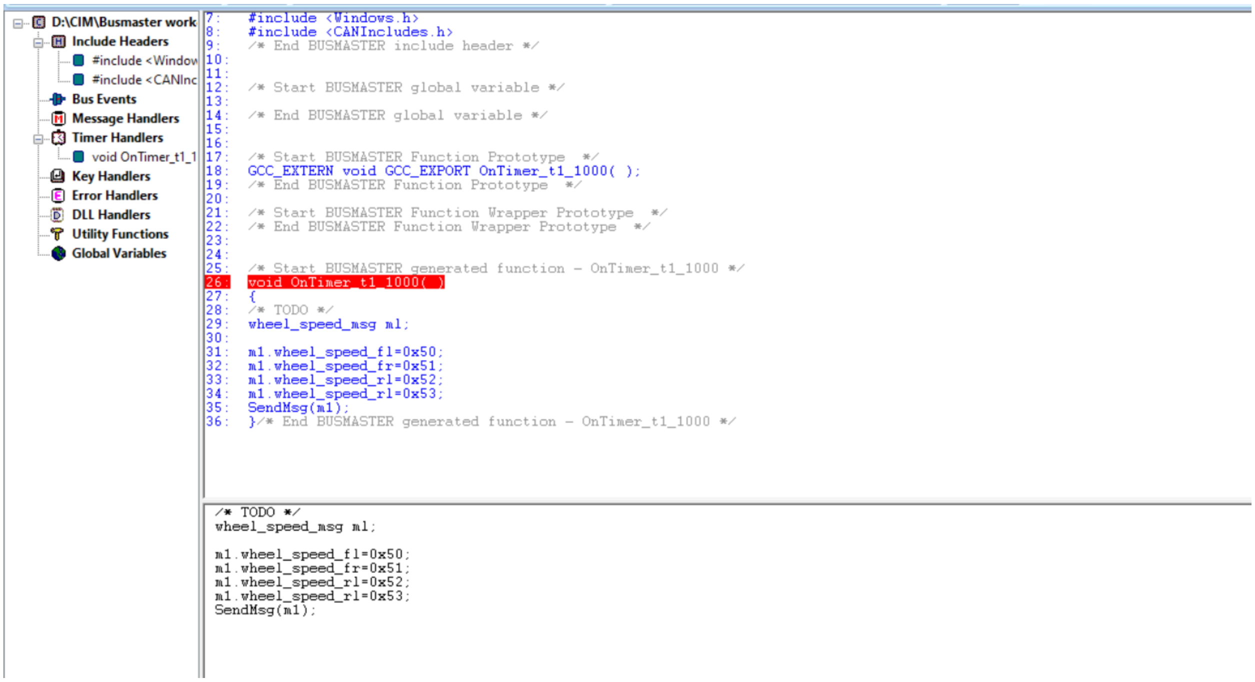
2)Creating Node:

Creating 3 nodes namely

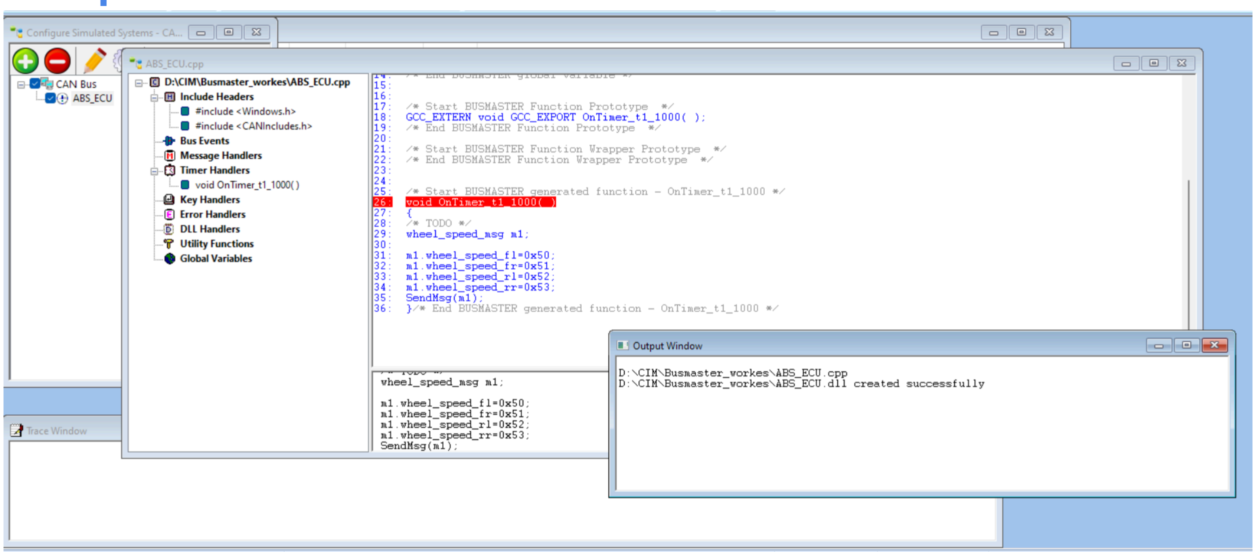
- ABS_ECU
- BMS_ECU
- dash_board_ECU

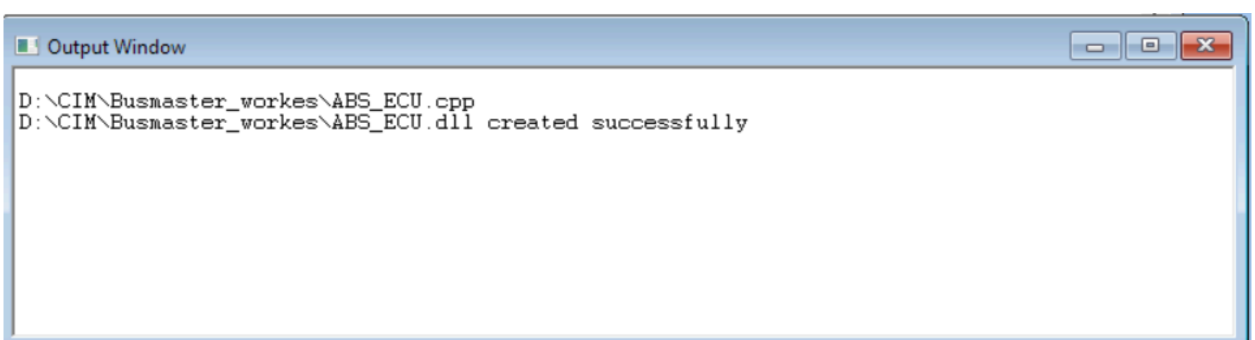
And writing code for timer handler & message handlers.

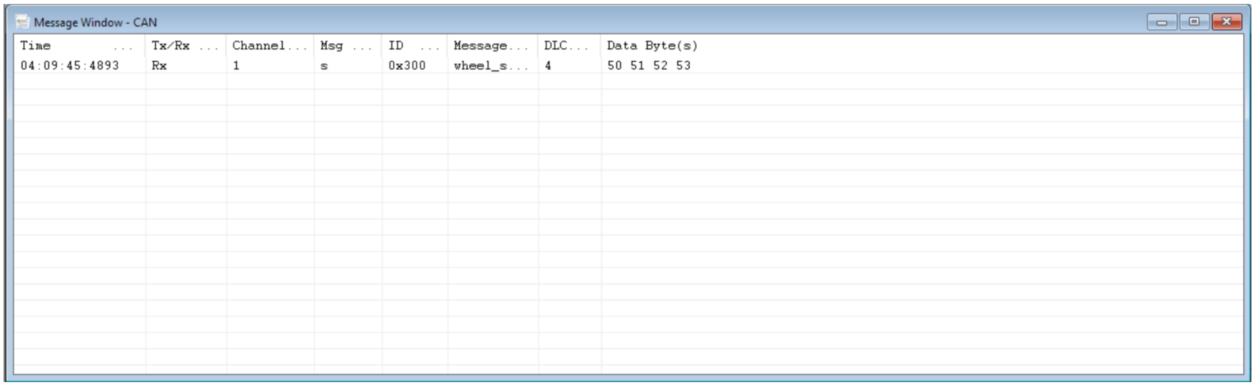

ABS_ECU.cpp
Define New Timer Handler
Timer Handler Name:
t1
Timer Value:
1000
msec
OK
Cancel
Error Handlers
DLL Handlers
Utility Functions
Global Variables
19: /* End BUSMASTER Function Prototype
21: /* Start BUSMASTER Function Wrapper
22: /* End BUSMASTER Function Wrapper
25: /* Start BUSMASTER generated function
26: void OnKey_b(unsigned char KeyValue)
27: {
28: /* TODO */
29: }/* End BUSMASTER generated function
/* TODO */


D:\CIM\Busmaster work
Include Headers
#include <Window
#include <CANInc
Bus Events
Message Handlers
Timer Handlers
void OnTimer_t1_1
Key Handlers
Error Handlers
DLL Handlers
Utility Functions
Global Variables
#include <Windows.h>
#include <CANIncludes.h>
/* End BUSMASTER include header */
/* Start BUSMASTER global variable */
/* End BUSMASTER global variable */
/* Start BUSMASTER Function Prototype */
GCC_EXTERN void GCC_EXPORT OnTimer_t1_1000( );
/* End BUSMASTER Function Prototype */
/* Start BUSMASTER Function Wrapper Prototype */
/* End BUSMASTER Function Wrapper Prototype */
/* Start BUSMASTER generated function - OnTimer_t1_1000 */
void OnTimer_t1_1000( )
{
/* TODO */
wheel_speed_msg m1;
m1.wheel_speed_fl=0x50;
m1.wheel_speed_fr=0x51;
m1.wheel_speed_rl=0x52;
m1.wheel_speed_rl=0x53;
SendMsg(m1);
}/* End BUSMASTER generated function - OnTimer_t1_1000 */
/* TODO */
wheel_speed_msg m1;
m1.wheel_speed_fl=0x50;
m1.wheel_speed_fr=0x51;
m1.wheel_speed_rl=0x52;
m1.wheel_speed_rl=0x53;
SendMsg(m1);

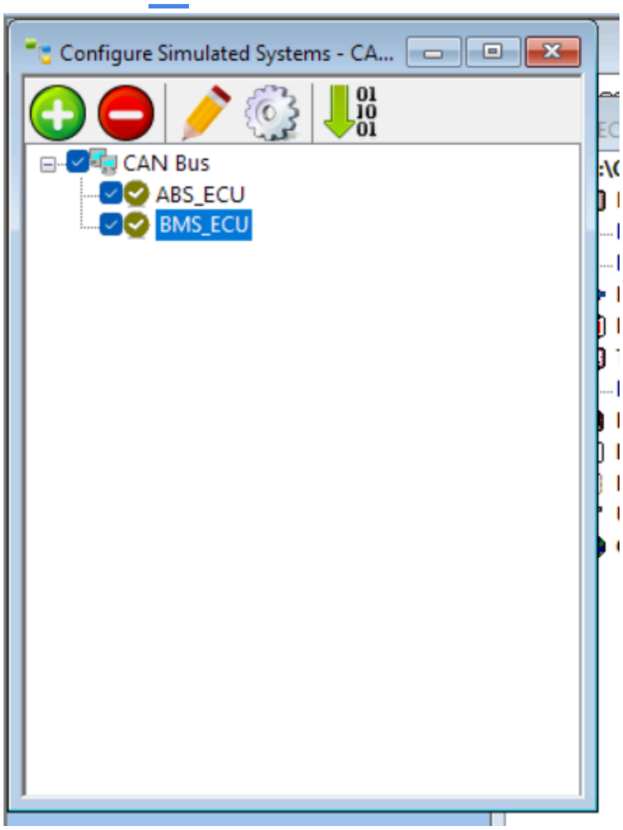
output:

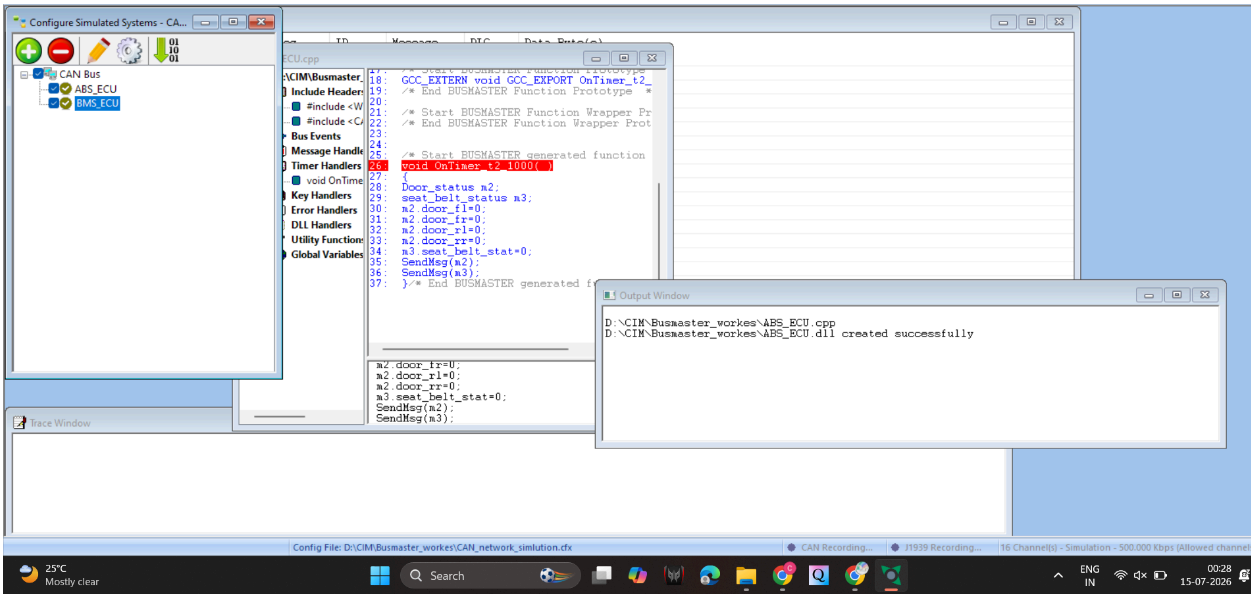

ABS_ECU.cpp
CAN Bus
ABS_ECU
D:\CIM\Busmaster_workes\ABS_ECU.cpp
Include Headers
Bus Events
Message Handlers
Timer Handlers
void OnTimer_t1_1000( )
Key Handlers
Error Handlers
DLL Handlers
Utility Functions
Global Variables
GCC_EXTERN void GCC_EXPORT OnTimer_t1_1000( );
void OnTimer_t1_1000( )
wheel_speed_msg m1;
m1.wheel_speed_fl=0x50;
m1.wheel_speed_fr=0x51;
m1.wheel_speed_rl=0x52;
m1.wheel_speed_rr=0x53;
SendMsg(m1);
Output Window
D:\CIM\Busmaster_workes\ABS_ECU.cpp
D:\CIM\Busmaster_workes\ABS_ECU.dll created successfully
Trace Window

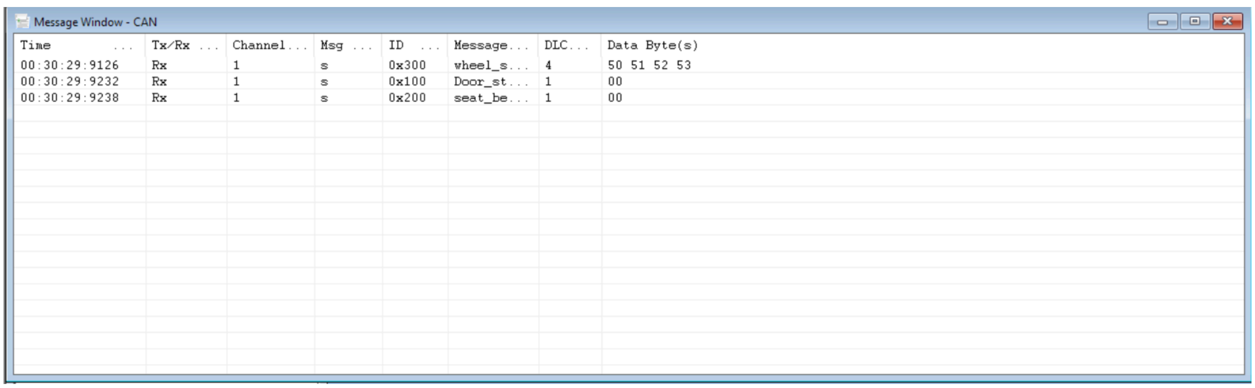

Output Window
D:\CIM\Busmaster_workes\ABS_ECU.cpp
D:\CIM\Busmaster_workes\ABS_ECU.dll created successfully

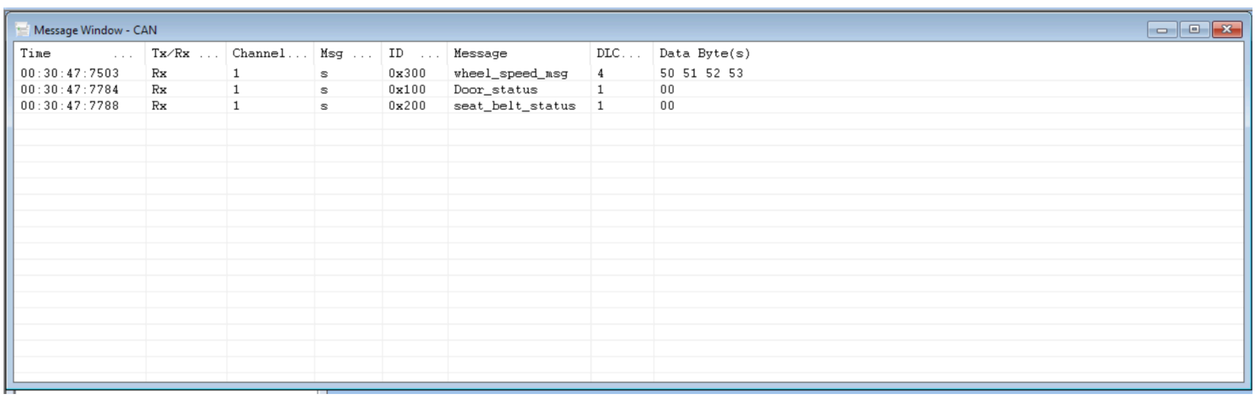

Message Window - CAN
Time ... Tx/Rx ... Channel... Msg ... ID ... Message... DLC... Data Byte(s)
04:09:45:4893 Rx 1 s 0x300 wheel_s... 4 50 51 52 53

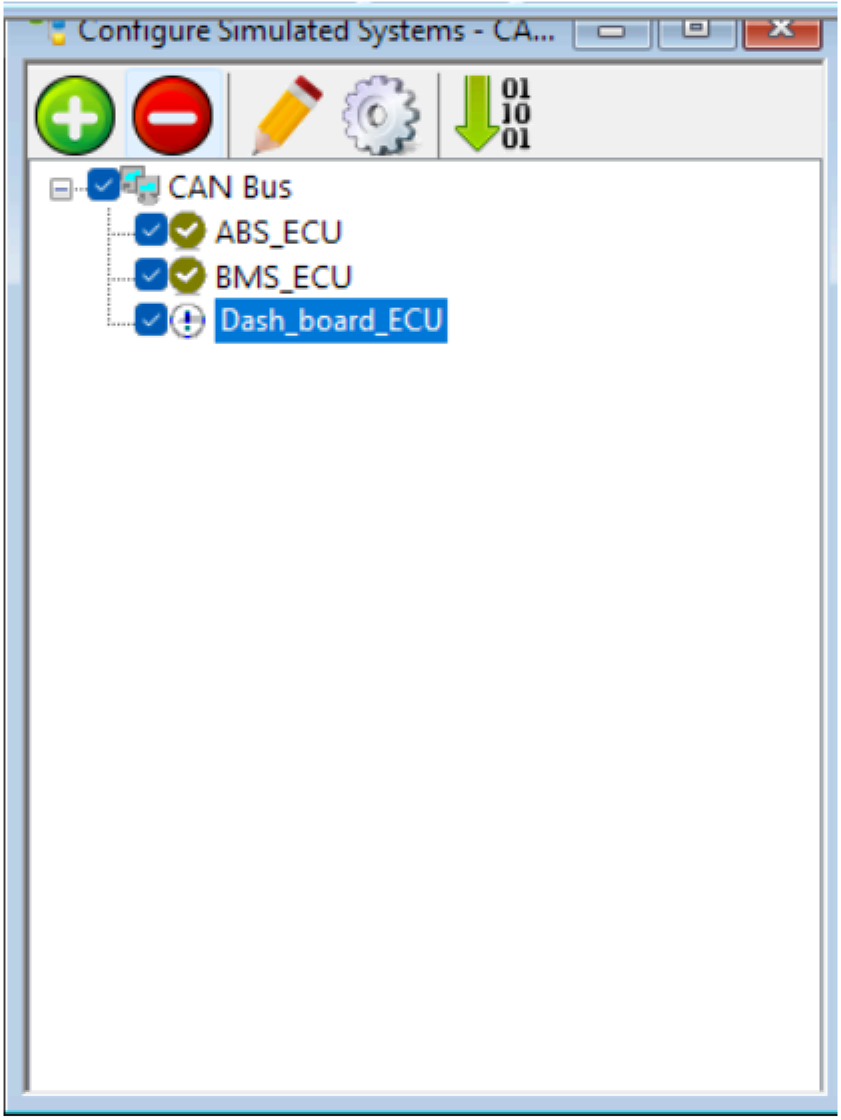
BMS_ECU:

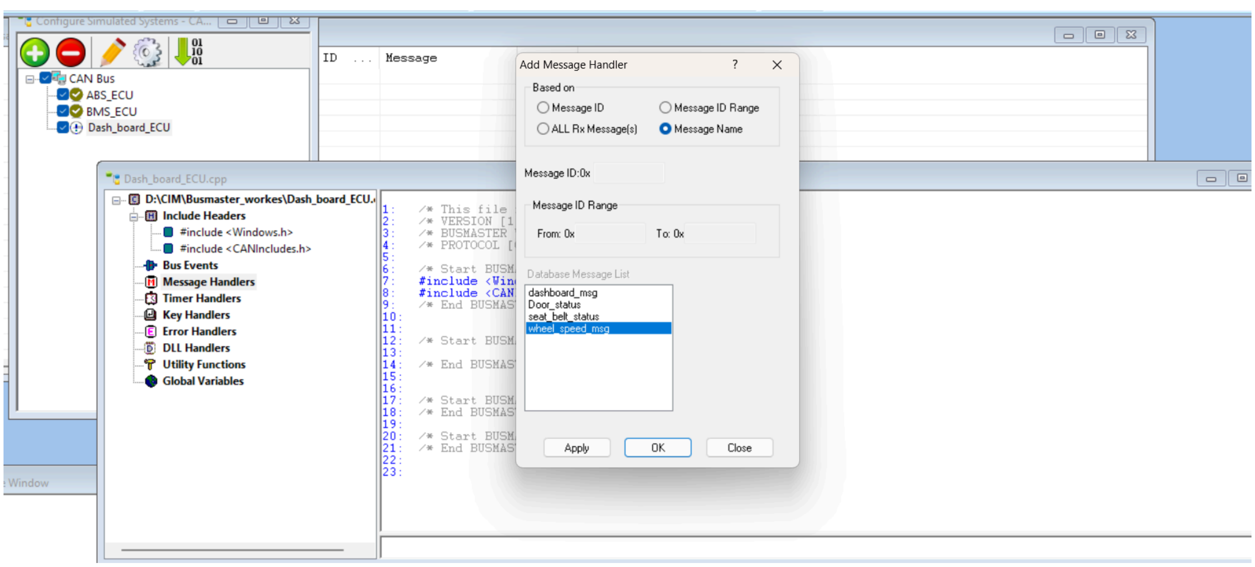

Configure Simulated Systems - CA...
01
10
01
CAN Bus
ABS_ECU
BMS_ECU

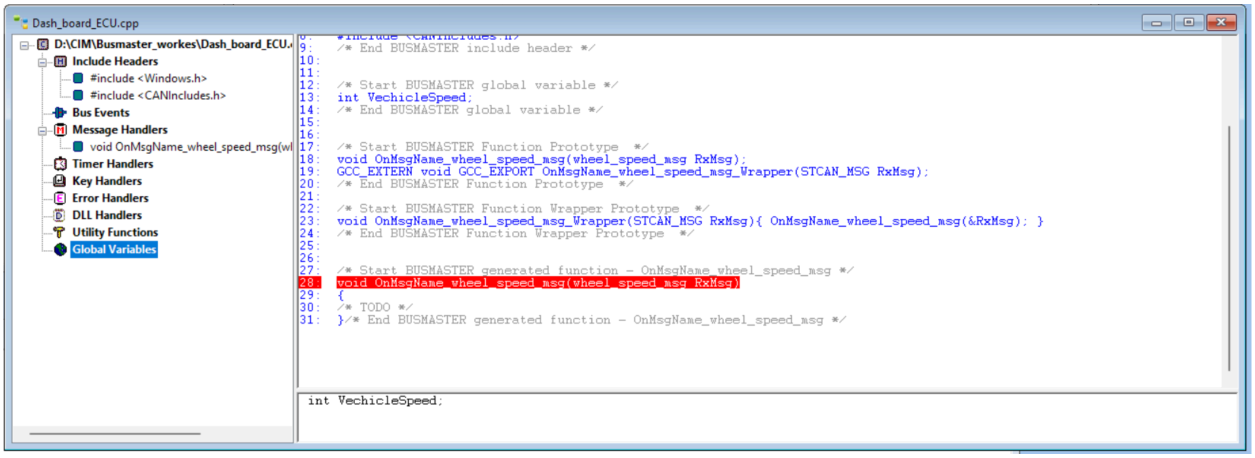

Configure Simulated Systems - CA...
CAN Bus
ABS_ECU
BMS_ECU
Include Headers
Bus Events
Timer Handlers
Key Handlers
Error Handlers
DLL Handlers
Global Variables
GCC_EXTERN void GCC_EXPORT OnTimer_t2_
/* End BUSMASTER Function Prototype
/* Start BUSMASTER Function Wrapper Pr
/* End BUSMASTER Function Wrapper Prot
/* Start BUSMASTER generated function
void OnTimer_t2_1000( )
{
Door_status m2;
seat_belt_status m3;
m2.door_fl=0;
m2.door_fr=0;
m2.door_rl=0;
m2.door_rr=0;
m3.seat_belt_stat=0;
SendMsg(m2);
SendMsg(m3);
}/* End BUSMASTER generated f
Output Window
D:\CIM\Busmaster_workes\ABS_ECU.cpp
D:\CIM\Busmaster_workes\ABS_ECU.dll created successfully
Trace Window
Config File: D:\CIM\Busmaster_workes\CAN_network_simlution.cfx
CAN Recording...
J1939 Recording...
25°C
Mostly clear
Search
ENG
IN
00:28
15-07-2026

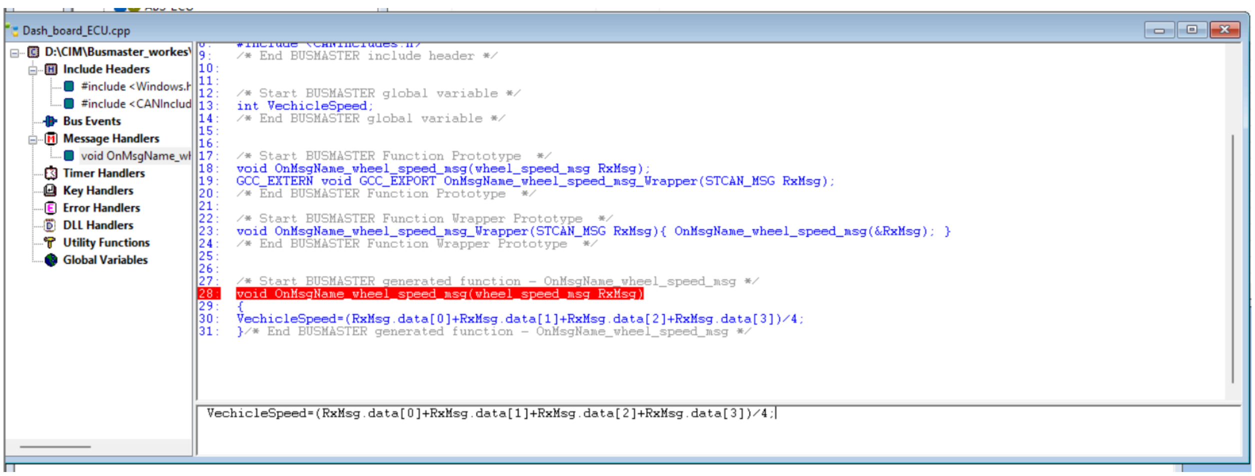

Message Window - CAN
Time ... Tx/Rx ... Channel... Msg ... ID ... Message... DLC... Data Byte(s)
00:30:29:9126 Rx 1 s 0x300 wheel_s... 4 50 51 52 53
00:30:29:9232 Rx 1 s 0x100 Door_st... 1 00
00:30:29:9238 Rx 1 s 0x200 seat_be... 1 00

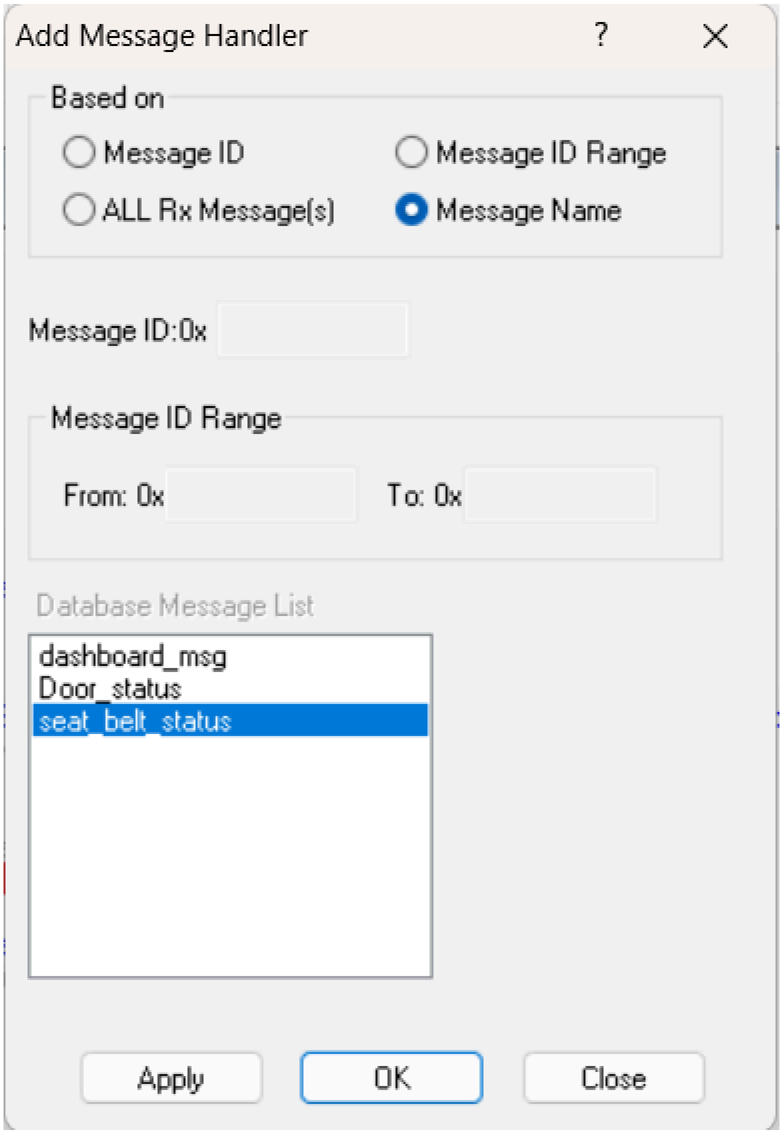

Message Window - CAN
Time ... Tx/Rx ... Channel... Msg ... ID ... Message DLC... Data Byte(s)
00:30:47:7503 Rx 1 s 0x300 wheel_speed_msg 4 50 51 52 53
00:30:47:7784 Rx 1 s 0x100 Door_status 1 00
00:30:47:7788 Rx 1 s 0x200 seat_belt_status 1 00

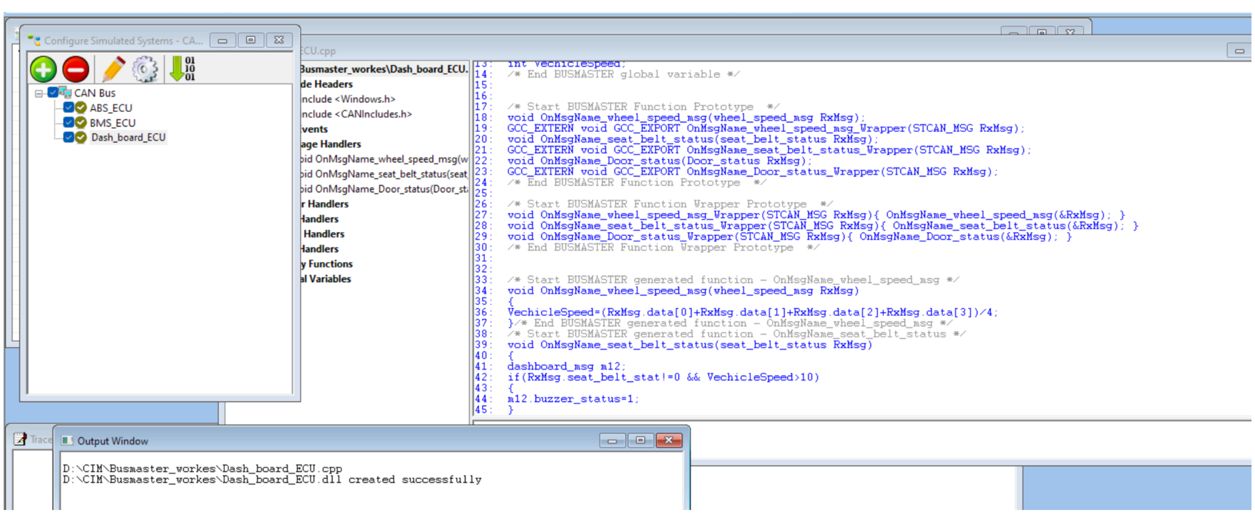

Configure Simulated Systems - CA...
01
10
01
CAN Bus
ABS_ECU
BMS_ECU
Dash_board_ECU

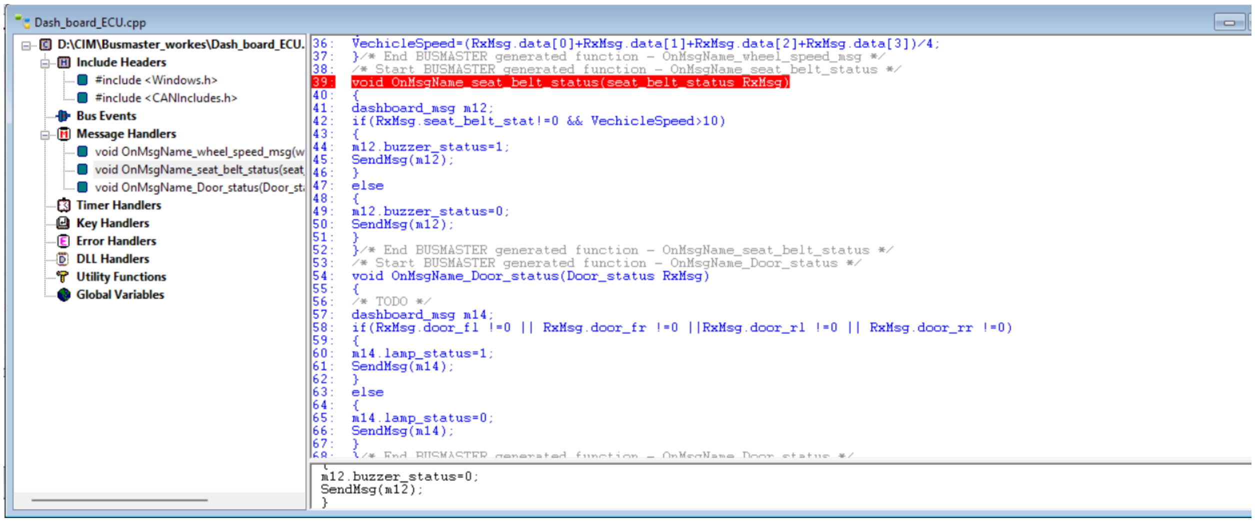

Configure Simulated Systems - CA...
CAN Bus
ABS_ECU
BMS_ECU
Dash_board_ECU
ID
Message
Add Message Handler
Based on
Message ID
Message ID Range
ALL Rx Message(s)
Message Name
Message ID:0x
Message ID Range
From: 0x
To: 0x
Database Message List
dashboard_msg
Door_status
seat_belt_status
wheel_speed_msg
Apply
OK
Close
Dash_board_ECU.cpp
Include Headers
#include <Windows.h>
#include <CANIncludes.h>
Bus Events
Message Handlers
Timer Handlers
Key Handlers
Error Handlers
DLL Handlers
Utility Functions
Global Variables


Dash_board_ECU.cpp
D:\CIM\Busmaster_workes\Dash_board_ECU.
Include Headers
#include <Windows.h>
#include <CANIncludes.h>
Bus Events
Message Handlers
void OnMsgName_wheel_speed_msg(wl
Timer Handlers
Key Handlers
Error Handlers
DLL Handlers
Utility Functions
Global Variables
/* End BUSMASTER include header */
/* Start BUSMASTER global variable */
int VechicleSpeed;
/* End BUSMASTER global variable */
/* Start BUSMASTER Function Prototype */
void OnMsgName_wheel_speed_msg(wheel_speed_msg RxMsg);
GCC_EXTERN void GCC_EXPORT OnMsgName_wheel_speed_msg_Wrapper(STCAN_MSG RxMsg);
/* End BUSMASTER Function Prototype */
/* Start BUSMASTER Function Wrapper Prototype */
void OnMsgName_wheel_speed_msg_Wrapper(STCAN_MSG RxMsg){ OnMsgName_wheel_speed_msg(&RxMsg); }
/* End BUSMASTER Function Wrapper Prototype */
/* Start BUSMASTER generated function - OnMsgName_wheel_speed_msg */
void OnMsgName_wheel_speed_msg(wheel_speed_msg RxMsg)
{
/* TODO */
}/* End BUSMASTER generated function - OnMsgName_wheel_speed_msg */
int VechicleSpeed;

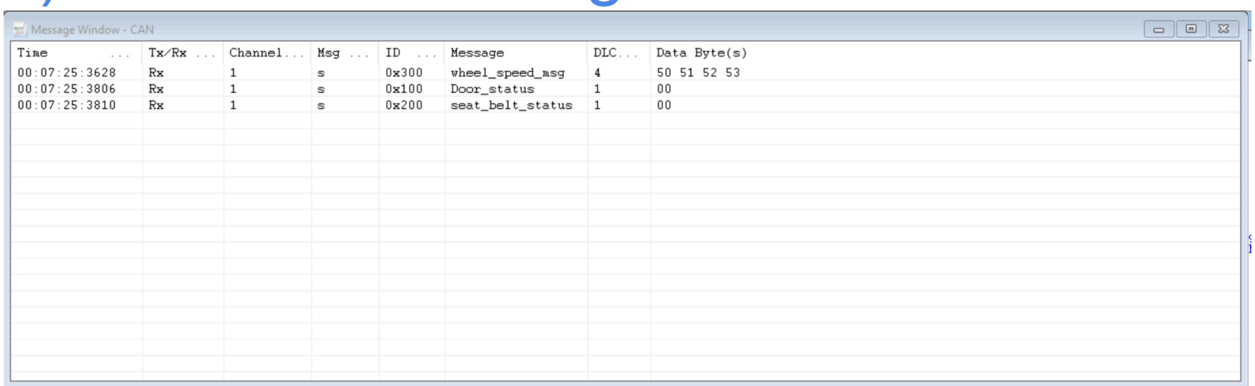

Dash_board_ECU.cpp
D:\CIM\Busmaster_workes\
Include Headers
#include <Windows.h
#include <CANInclud
Bus Events
Message Handlers
void OnMsgName_wl
Timer Handlers
Key Handlers
Error Handlers
DLL Handlers
Utility Functions
Global Variables
/* End BUSMASTER include header */
/* Start BUSMASTER global variable */
int VechicleSpeed;
/* End BUSMASTER global variable */
/* Start BUSMASTER Function Prototype */
void OnMsgName_wheel_speed_msg(wheel_speed_msg RxMsg);
GCC_EXTERN void GCC_EXPORT OnMsgName_wheel_speed_msg_Wrapper(STCAN_MSG RxMsg);
/* End BUSMASTER Function Prototype */
/* Start BUSMASTER Function Wrapper Prototype */
void OnMsgName_wheel_speed_msg_Wrapper(STCAN_MSG RxMsg){ OnMsgName_wheel_speed_msg(&RxMsg); }
/* End BUSMASTER Function Wrapper Prototype */
/* Start BUSMASTER generated function - OnMsgName_wheel_speed_msg */
void OnMsgName_wheel_speed_msg(wheel_speed_msg RxMsg)
{
VechicleSpeed=(RxMsg.data[0]+RxMsg.data[1]+RxMsg.data[2]+RxMsg.data[3])/4;
}/* End BUSMASTER generated function - OnMsgName_wheel_speed_msg */
VechicleSpeed=(RxMsg.data[0]+RxMsg.data[1]+RxMsg.data[2]+RxMsg.data[3])/4;


Add Message Handler
?
Based on
Message ID
Message ID Range
ALL Rx Message(s)
Message Name
Message ID:0x
Message ID Range
From: 0x
To: 0x
Database Message List
dashboard_msg
Door_status
seat_belt_status
Apply
OK
Close


Configure Simulated Systems - CA...
CAN Bus
ABS_ECU
BMS_ECU
Dash_board_ECU
/* End BUSMASTER global variable */
/* Start BUSMASTER Function Prototype */
void OnMsgName_wheel_speed_msg(wheel_speed_msg RxMsg);
GCC_EXTERN void GCC_EXPORT OnMsgName_wheel_speed_msg_Wrapper(STCAN_MSG RxMsg);
void OnMsgName_seat_belt_status(seat_belt_status RxMsg);
GCC_EXTERN void GCC_EXPORT OnMsgName_seat_belt_status_Wrapper(STCAN_MSG RxMsg);
void OnMsgName_Door_status(Door_status RxMsg);
GCC_EXTERN void GCC_EXPORT OnMsgName_Door_status_Wrapper(STCAN_MSG RxMsg);
/* End BUSMASTER Function Prototype */
/* Start BUSMASTER Function Wrapper Prototype */
void OnMsgName_wheel_speed_msg_Wrapper(STCAN_MSG RxMsg){ OnMsgName_wheel_speed_msg(&RxMsg); }
void OnMsgName_seat_belt_status_Wrapper(STCAN_MSG RxMsg){ OnMsgName_seat_belt_status(&RxMsg); }
void OnMsgName_Door_status_Wrapper(STCAN_MSG RxMsg){ OnMsgName_Door_status(&RxMsg); }
/* End BUSMASTER Function Wrapper Prototype */
/* Start BUSMASTER generated function - OnMsgName_wheel_speed_msg */
void OnMsgName_wheel_speed_msg(wheel_speed_msg RxMsg)
{
VechicleSpeed=(RxMsg.data[0]+RxMsg.data[1]+RxMsg.data[2]+RxMsg.data[3])/4;
}/* End BUSMASTER generated function - OnMsgName_wheel_speed_msg */
/* Start BUSMASTER generated function - OnMsgName_seat_belt_status */
void OnMsgName_seat_belt_status(seat_belt_status RxMsg)
{
dashboard_msg m12;
if(RxMsg.seat_belt_stat!=0 && VechicleSpeed>10)
{
m12.buzzer_status=1;
}
Output Window
D:\CIM\Busmaster_workes\Dash_board_ECU.cpp
D:\CIM\Busmaster_workes\Dash_board_ECU.dll created successfully

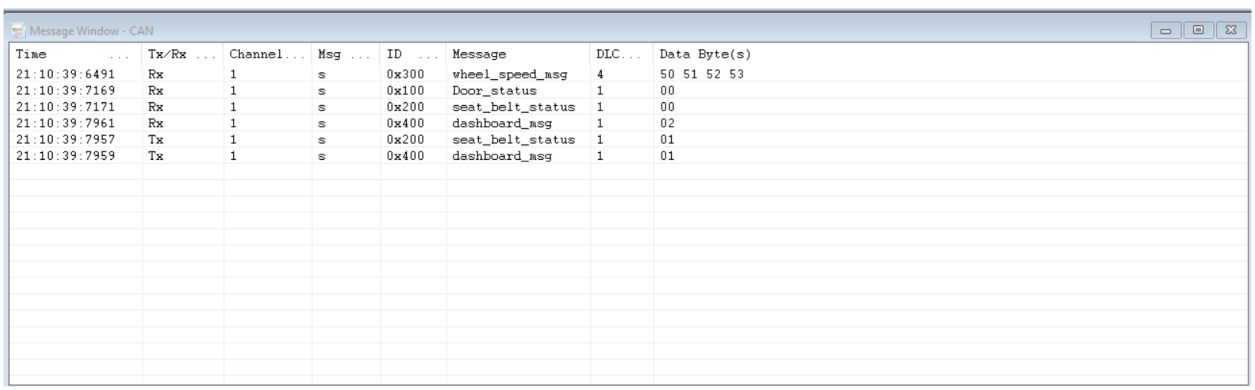


3) Data Transmitting from the dataBase:



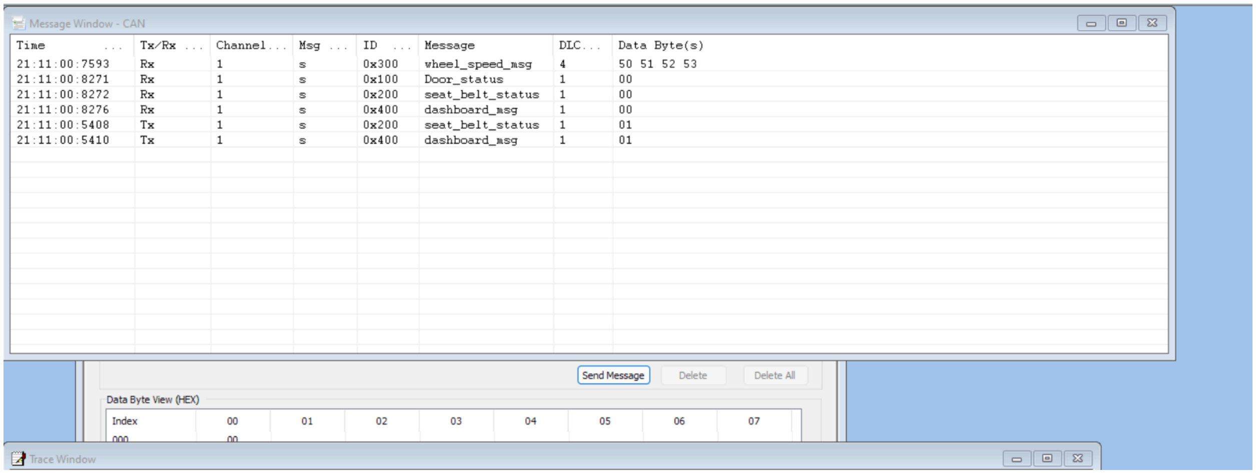
Message Window - CAN

Time	Tx/Rx	Channel	Msg	ID	Message	DLC	Data Byte(s)
00:07:25:3628	Rx	1	s	0x300	wheel_speed_msg	4	50 51 52 53
00:07:25:3806	Rx	1	s	0x100	Door_status	1	00
00:07:25:3810	Rx	1	s	0x200	seat_belt_status	1	00



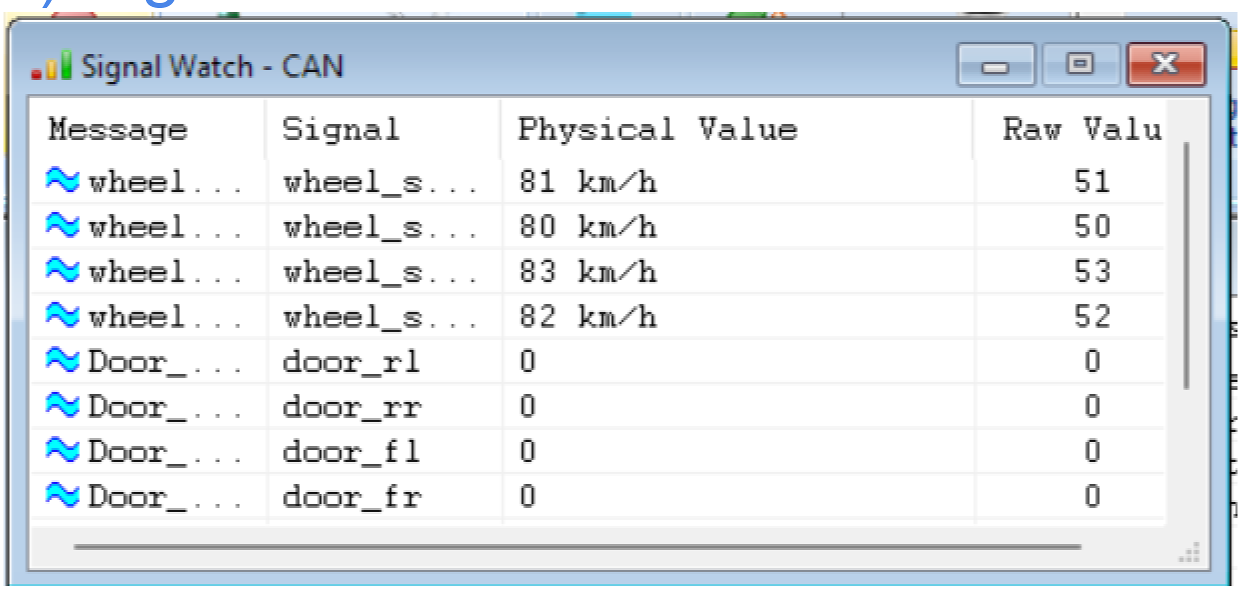
Message Window - CAN

Time	Tx/Rx	Channel	Msg	ID	Message	DLC	Data Byte(s)
21:10:39:6491	Rx	1	s	0x300	wheel_speed_msg	4	50 51 52 53
21:10:39:7169	Rx	1	s	0x100	Door_status	1	00
21:10:39:7171	Rx	1	s	0x200	seat_belt_status	1	00
21:10:39:7961	Rx	1	s	0x400	dashboard_msg	1	02
21:10:39:7957	Tx	1	s	0x200	seat_belt_status	1	01
21:10:39:7959	Tx	1	s	0x400	dashboard_msg	1	01



Time	Tx/Rx	Channel	Msg	ID	Message	DLC	Data Byte(s)
21:11:00:7593	Rx	1	s	0x300	wheel_speed_msg	4	50 51 52 53
21:11:00:8271	Rx	1	s	0x100	Door_status	1	00
21:11:00:8272	Rx	1	s	0x200	seat_belt_status	1	00
21:11:00:8276	Rx	1	s	0x400	dashboard_msg	1	00
21:11:00:5408	Tx	1	s	0x200	seat_belt_status	1	01
21:11:00:5410	Tx	1	s	0x400	dashboard_msg	1	01

4) Signal Watch from Database values:



Message	Signal	Physical Value	Raw Valu
wheel...	wheel_s...	81 km/h	51
wheel...	wheel_s...	80 km/h	50
wheel...	wheel_s...	83 km/h	53
wheel...	wheel_s...	82 km/h	52
Door_...	door_rl	0	0
Door_...	door_rr	0	0
Door_...	door_fl	0	0
Door_...	door_fr	0	0

5) Signal Graph: